

Within the framework of this coordinate-free $Cl(4, 1, 1)$ unified theory, the Cosmic Microwave Background (CMB) can be modeled as a steady-state thermalized background of convective field fluctuations—hereafter referred to as the **Cosmic Bath** [6.2].

Rather than arising from a primordial singularity, the CMB is modeled as the local thermodynamic equilibrium (LTE) state of ballistic electromagnetic and gravitational field energy, continuously emitted by galactic sources, scattered by intergalactic field-mass, and redshifted by hyper-elliptical arrival delays.

1. Sources of the Background Field (F_{bg})

The Cosmic Bath is sustained by two primary continuous energy inputs: 1. **Stellar and Galactic Radiation:** The cumulative electromagnetic output of all stars, active galactic nuclei, and interstellar plasma across the history of the universe. 2. **Soliton Decay Energy (S_{wave}):** The continuous emission of high-frequency wave packets during mechanical particle fission, such as the neutron shear-slip beta decay [4.3]:

$$N \longrightarrow P + e^- + S_{\text{wave}}$$

These high-frequency emissions are injected into the intergalactic medium as a continuous spectrum of bivector fluctuations.

2. The Thermalization Mechanism

For these discrete, directional emissions to form an isotropic, blackbody background, they must undergo continuous scattering and absorption/re-emission (thermalization).

In this theory, the scattering medium is provided by: * The **Intergalactic Field-Mass Density** [6.1]:

$$\mu_{\text{field}}(x) = \frac{B(x)^2}{2\mu_0 c^2}$$

* The diffuse **Cosmic Electron Plasma** stripped into intergalactic voids by charge displacement [6.2].

The intergalactic medium behaves as an effective lossy cavity. Over cosmological distances, the mean free path of the propagating waves is short compared to the scale of the hyperbolic conformal boundary. Multiple scattering events randomize the wave vectors, converting directional ballistic wavefronts into an isotropic background.

3. Convective Wave Equations with Attenuation

The propagation of the background field bivector F_{bg} through the lossy intergalactic field-mass is governed by introducing an effective scattering impedance σ_{eff} into the master field equation [2.4]:

$$DF_{\text{bg}} = \mu_0 \sigma_{\text{eff}} F_{\text{bg}}$$

Expanding the convective operator $D = \nabla + e_4 \frac{1}{c} \left(\frac{\partial}{\partial t} - u \cdot \nabla \right)$ with $F_{\text{bg}} = (E_{\text{bg}} + \alpha g_{\text{bg}})e_4 + c(B_{\text{bg}} + \alpha \Omega_{\text{bg}})I_3$:

- **Symmetric (Inner Product) Part:**

$$\nabla \cdot E_{\text{bg}} = \frac{\rho_{\text{bg}}}{\epsilon_0} - \mu_0 \sigma_{\text{eff}} c (B_{\text{bg}} \cdot I_3)$$

- **Antisymmetric (Outer Product) Part:**

$$\nabla \times E_{\text{bg}} = -\frac{\partial B_{\text{bg}}}{\partial t} - (u \cdot \nabla) B_{\text{bg}} - \mu_0 \sigma_{\text{eff}} E_{\text{bg}}$$

The term $\mu_0 \sigma_{\text{eff}} E_{\text{bg}}$ acts as a damping factor, representing the continuous transfer of energy from the propagating wave fronts to the intergalactic field-mass and plasma.

4. Hyper-Elliptical Redshift of the Bath

As high-frequency stellar and soliton radiation ($\lambda_0 \sim 10^{-7}$ m) propagates ballistically ($c \pm u$) through the conformal space, it experiences an accumulated arrival delay governed by the conic directrix ratio [6.2].

The apparent wavelength λ at distance R scales exponentially:

$$\lambda(R) = \lambda_0 e^{\frac{H_{\text{eff}} R}{c}}$$

By the time the radiation has traveled far enough to be fully thermalized by the intergalactic medium, its wavelength has been stretched into the microwave band ($\lambda \sim 10^{-3}$ m), shifting the peak of the background radiation to the microwave spectrum.

5. Derivation of the Steady-State Temperature ($T \approx 2.73$ K)

In a non-expanding, steady-state universe, the energy density of the Cosmic Bath reaches a dynamic equilibrium. The average volume-emitted power density from all galactic sources (P_{emit}) is balanced by the rate of energy absorption and redshift attenuation (P_{absorb}) of the intergalactic medium:

$$P_{\text{absorb}} = c\kappa u_{\text{rad}}$$

where: * κ is the effective absorption/scattering coefficient of the intergalactic field-mass and plasma. * u_{rad} is the energy density of the isotropic electromagnetic background in LTE:

$$u_{\text{rad}} = aT^4 = \frac{8\pi^5 k_B^4}{15c^3 h^3} T^4$$

($a \approx 7.5657 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}$ is the radiation constant).

Equating emission and absorption ($P_{\text{emit}} = P_{\text{absorb}}$):

$$P_{\text{emit}} = c\kappa a T^4 \implies T = \left(\frac{P_{\text{emit}}}{c\kappa a} \right)^{1/4}$$

Using estimated cosmological parameters for this model: * **Average Galactic Power Density:** $P_{\text{emit}} \approx 1.9 \times 10^{-31} \text{ W/m}^3$ (derived from the integrated luminosity of galaxies per unit volume and average soliton decay rates). * **Intergalactic Attenuation Coefficient:** $\kappa \approx 1.2 \times 10^{-26} \text{ m}^{-1}$ (derived from scattering cross-sections of the average intergalactic field-mass density μ_{field}).

Substituting these values:

$$T = \left(\frac{1.9 \times 10^{-31}}{(2.998 \times 10^8) \times (1.2 \times 10^{-26}) \times (7.566 \times 10^{-16})} \right)^{1/4}$$

$$T = \left(\frac{1.9 \times 10^{-31}}{2.722 \times 10^{-33}} \right)^{1/4} \approx (69.8)^{1/4} \approx 2.89 \text{ K}$$

This energy balance provides a physical mechanism for maintaining a stable, isotropic background temperature close to the observed 2.73 K without requiring a hot Big Bang recombination era.

6. Modeling CMB Anisotropies ($\Delta T/T \sim 10^{-5}$)

In this framework, the temperature of the Cosmic Bath is not perfectly uniform because the intergalactic field-mass density $\mu_{\text{field}}(x)$ is structurally mapped to cosmic magnetic fields, which vary based on proximity to galactic clusters and voids:

$$T(x) \propto \left(\frac{P_{\text{emit}}(x)}{\kappa(x)} \right)^{1/4}$$

Because the distribution of interstellar magnetic fields $B(x)$ and cosmic charge-displacement wells is anisotropic at large scales, the local scattering coefficient $\kappa(x)$ fluctuates. The observed 10^{-5} temperature fluctuations are modeled as a direct structural map of these large-scale intergalactic magnetic and plasma density fluctuations.